Question	Answer	Marks
1(a)	force per unit mass	**B1**
1(b)(i)	radial	**B1**
	towards (centre of) planet	**B1**
1(b)(ii)	(changes in) height (very) much smaller than radius	**B1**
	(radius + height)^2 ≈ radius^2 **or** field lines are approximately parallel	**B1**
1(c)(i)	$9.81 \times R^2 = 3.99 \times 10^{14}$	**C1**
	$R = 6.38 \times 10^6$ m	**A1**
1(c)(ii)	gravitational potential $= -(GM/R)$ $= -(3.99 \times 10^{14}) / (6.38 \times 10^6)$	**C1**
	$= -6.25 \times 10^7$ J kg^{-1}	**A1**
1(d)	potential (energy) zero at infinite separation	**B1**
	(gravitational) force is attractive	**B1**

Question	Answer	Marks